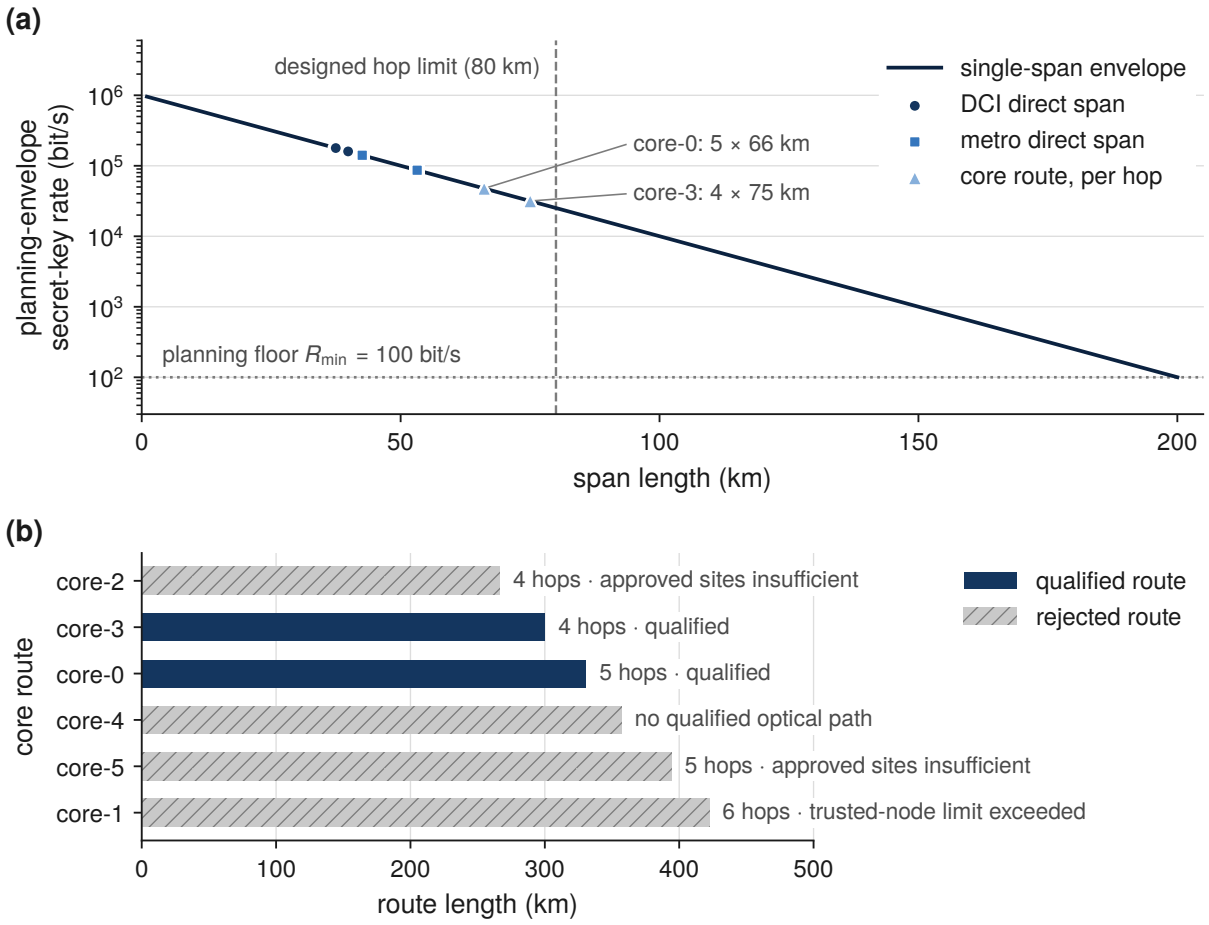
(a)
designed hop limit (80 km)
single-span envelope
DCI direct span
metro direct span
core route, per hop
core-0: 5 × 66 km
core-3: 4 × 75 km
planning-envelope secret-key rate (bit/s)
10^6
10^5
10^4
10^3
10^2
planning floor R_{min} = 100 bit/s
0
50
100
150
200
span length (km)
(b)
core route
core-2
core-3
core-0
core-4
core-5
core-1
4 hops · approved sites insufficient
4 hops · qualified
5 hops · qualified
no qualified optical path
5 hops · approved sites insufficient
6 hops · trusted-node limit exceeded
qualified route
rejected route
0
100
200
300
400
500
route length (km)